

외계행성 대기

Exoplanetary Atmospheres

Lecture 4: Radiative Transfer II — The Transfer Equation, Temperatures, and Albedos 2026 September 28 (Monday)

Based on: Chapters 3 and 5 of Seager, *Exoplanet Atmospheres* (2010), and Chapter 2 of Heng, *Exoplanetary Atmospheres: Theoretical Concepts and Foundations* (2017)

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Outline – Lecture 4

Optical Depth and the Mean Free Path

Local Thermodynamic Equilibrium and the Source Function

The Equation of Radiative Transfer

Energy Balance

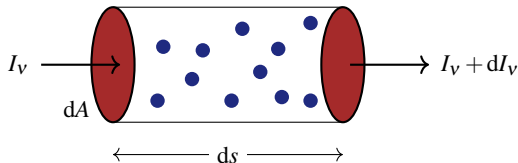
Planetary Temperatures

Planetary Albedos

Summary

Absorption (Extinction)

If the radiation travels through a medium that absorbs radiation, the energy in the beam is reduced:



- ▶ Let n denote the number density of absorbers (particles per unit volume).
- ▶ Assume that each absorber has a cross-sectional area σ_V (in units of cm^2).

‘geometric’ cross section: $\sigma = \pi r^2$ for a spherical particle of radius r

- ▶ If the beam travels through ds , the total area of the absorbers is

$$\text{number of absorbers} \times \text{cross section} = (n \times dA \times ds) \times \sigma_V.$$

Absorption coefficient

Fraction of radiation absorbed = fraction of area blocked:

$$\frac{dI_V}{I_V} = -\frac{n dA ds \sigma_V}{dA} = -n \sigma_V ds,$$

$\Rightarrow$

$$\frac{dI_V}{ds} = -\alpha_V I_V$$

$$dI_V = -n \sigma_V I_V ds \equiv -\alpha_V I_V ds$$

- The *absorption coefficient* is defined as $\alpha_V \equiv n \sigma_V$ (units: cm^{-1}), meaning the *total cross-sectional area per unit volume*.

$$\begin{aligned}\alpha_V &= n \sigma_V \quad [\text{cm}^{-1}] \\ &= \rho \kappa_V\end{aligned}$$

where ρ is the mass density and κ_V is called the **mass absorption coefficient** or the **opacity coefficient** [$\text{cm}^2 \text{g}^{-1}$].

Notation: α_V is the per-length coefficient that Seager calls κ (Heng's α_e); our $\kappa_V = \alpha_V / \rho$ is Seager's κ_m . From here on we follow Seager and write κ for the per-length coefficient.

The mean free path

The fraction of area blocked in ds , $n\sigma ds$, is also the *probability* that a single photon is stopped there. A photon therefore travels, on average, the distance over which that probability adds up to one. This is the **mean free path**, the mean distance a photon travels before it is stopped by an interaction with a molecule or other atmospheric particle:

$$l = \frac{1}{n\sigma} = \frac{1}{\alpha}, \quad d\tau = \frac{ds}{l}$$

- ▶ The optical depth counts the path length in units of mean free paths; $\tau \sim 1$ means “about one interaction”, as Heng’s intuitive argument said.
- ▶ Typical values: in Earth’s lower atmosphere at visible wavelengths (Rayleigh scattering, $\sigma \simeq 5 \times 10^{-27} \text{ cm}^2$, $n \simeq 2.5 \times 10^{19} \text{ cm}^{-3}$) $l \simeq 80 \text{ km}$, so the clear sky has $\tau \simeq 0.1$ vertically; inside a water cloud l is tens of meters; in the $15 \mu\text{m}$ CO_2 band core l is meters.

Optical depth, formally, and the plane-parallel convention

From our viewpoint distance is measured in km; for photons the natural distance scale is the optical depth, a measure of transparency. Along a path s , the optical depth τ is defined as the integral of the absorption coefficient:

$$d\tau(s, \nu) = -\alpha(s, \nu) ds$$

dimensionless; $\tau \ll 1$ optically thin (a photon typically crosses without interacting), $\tau > 1$ optically thick. Because α depends on frequency and position, so does $\tau(\mathbf{x}, \nu)$.

- In **plane-parallel** atmospheres τ is measured *backward* along the outgoing ray: $\tau = 0$ at the top of the atmosphere, increasing downward, because the observer looks down into the atmosphere. Hence the minus sign:

$$\tau(z, \nu) = - \int_{z_{\max}}^z \alpha(z', \nu) dz' = \int_z^{z_{\max}} \alpha(z', \nu) dz'.$$

We write τ_ν with the z dependence implied.

- Later we will see that τ is the dimensionless *e-folding* scale for the attenuation of hot radiation through a cooler gas layer.
- Seager's κ is Heng's α_e ; in mass units, $d\tau = \kappa_m \rho ds$, and with hydrostatic equilibrium $d\tau = \kappa_m dP/g$.

The mean free path, more formally

Start from $dI = -\alpha I dz$, i.e.

$$\frac{dI}{I} = -\frac{dz}{l} \quad \Rightarrow \quad I = I_0 e^{-z/l}.$$

The probability that a photon is absorbed between z and $z + dz$ is the fraction of the initial beam removed there,

$$dP(z) = \frac{I(z) - I(z + dz)}{I_0} = \frac{1}{l} e^{-z/l} dz \quad \Rightarrow \quad dP = e^{-\tau} d\tau,$$

a properly normalized probability distribution ($\int_0^\infty dP = 1$). Its expectation value is

$$\langle z \rangle = \int_0^\infty z dP(z) = \int_0^\infty \frac{z}{l} e^{-z/l} dz = l \quad \Rightarrow \quad \langle \tau \rangle = 1$$

The mean distance traveled before an interaction is indeed $l = 1/\alpha$, which justifies the name.

- In optical-depth units the distribution is $dP = e^{-\tau} d\tau$, with mean $\langle \tau \rangle = 1$ and standard deviation 1: interaction distances are broadly distributed, and a sizable fraction of photons travel $\tau > 2$ or 3 without interacting ($e^{-2} = 14\%$, $e^{-3} = 5\%$).
- Monte Carlo radiative transfer samples exactly this distribution: $\tau = -\ln \xi$ with ξ a uniform random number in $(0, 1]$, then the photon is moved to the point where the accumulated optical depth along its direction equals τ .

Local thermodynamic equilibrium (LTE)

Complete thermodynamic equilibrium requires thermal, chemical, and mechanical equilibrium — impossible across an atmosphere with huge gradients of T and P and an open boundary at the top. **Local thermodynamic equilibrium (LTE)** applies in a local region where the gradients are small compared to the photon mean free path.

- ▶ In LTE we assume all the conditions of thermodynamic equilibrium hold for the *matter*, but we let the *radiation field* depart from a blackbody. The radiation field in a planetary atmosphere is very different from B ; it must be computed from the transfer equation.
- ▶ Why it matters: the coefficients κ and ϵ are sums of number densities times cross sections over all **level populations** (the numbers of atoms or molecules in each quantum state). We need (1) the overall number densities and (2) the fraction in each state.
- ▶ The heart of the problem: the level populations are determined by the radiation field (photoexcitation, stimulated emission, photoionization) — the very quantity we are solving for. A full simultaneous solution of the radiation field and the populations is possible in principle but onerous in practice.
- ▶ **LTE decouples them:** all properties of the matter depend only on the local kinetic temperature and one other variable (ρ or P). Level populations follow the Boltzmann distribution, ionization the Saha equation, molecular abundances chemical equilibrium (Chapter 8 and Heng Chapter 7).

When is LTE valid?

LTE implies a strict connection between the matter and the local temperature; it is valid where that connection is physically realized.

- ▶ **High densities: collisions dominate radiative processes.** An electron in an atom or molecule is excited to a higher level by absorbing a photon. Later, because of a collision, it cascades downward, releasing the energy as kinetic energy of the colliding particles; after further elastic collisions this energy ends up in the thermal pool of the gas. In this case of collisional deexcitation the photon is said to be *destroyed* and converted into kinetic energy. In this way collisions couple the photons (the radiation field) to the matter temperature via the kinetic energy of the gas. (As long as the colliding particles have a Maxwellian velocity distribution, the level populations have their Boltzmann equilibrium values.)
- ▶ **A blackbody radiation field.** If the mean radiation field is a blackbody, $J(\mathbf{x}, \nu) = B(\mathbf{x}, \nu)$, the radiative rate equations that control the level populations also give an equilibrium (Boltzmann) distribution (Section 8.2.4). This second situation is not really relevant for planetary atmospheres, and from the first example it is clearly not a *necessary* condition for LTE.
- ▶ The criterion is a comparison of rates: LTE holds for a transition when the collisional deexcitation rate exceeds the radiative one, $n_{\text{coll}} q_{ul} \gg A_{ul}$, i.e. above a *critical density* that depends on the transition (Chapter 8).

LTE in Earth's atmosphere and in exoplanets

- ▶ **Earth:** LTE is actually valid from the ground up to about 60 km, largely because collisional rates between molecules dominate radiative rates, driving a Boltzmann distribution. We know from spectral measurements at the top of the atmosphere and at different altitudes that Earth's radiation field is *not* a blackbody — LTE and blackbody radiation are different things. Above ~ 60 km (mesosphere, thermosphere) the CO_2 $15\text{ }\mu\text{m}$ band and other bands are in non-LTE: collisions are too rare to keep the vibrational levels thermalized.
- ▶ **Exoplanets:** LTE is commonly used in exoplanet atmosphere calculations, and it is a good approximation for the pressures probed by emission spectra ($\gtrsim 1$ mbar). But there are regions where it is not valid: the very upper layers where radiation escapes freely through the open boundary. There the radiation field is very different from a blackbody *and* collisions do not dominate over radiative transition rates, causing a departure from LTE. Examples: CH_4 fluorescence at $3.3\text{ }\mu\text{m}$ in giant planets, the escaping upper atmospheres probed by UV transits, and the thermospheres heated by stellar XUV.
- ▶ **Validation:** for a given situation one could compare radiative and collisional rates, but to accurately validate LTE a non-LTE calculation itself may be necessary. In practice one checks whether the pressures probed by the observation are above the critical densities of the relevant transitions.

LTE and Kirchhoff's law

Kirchhoff's law of radiation states that in thermal equilibrium the emissivity of a body equals its absorptivity:

$$\epsilon_{\text{therm}}(\mathbf{x}, \nu) = \alpha_{\text{abs}}(\mathbf{x}, \nu) B(\mathbf{x}, \nu)$$

- ▶ The thermal emission at $\mathbf{x}$ is blackbody radiation weighted by the absorptive properties of the gas: any time a photon is absorbed by gas in LTE, the energy is reemitted as radiation, with amount and wavelength set by the Planck function times the absorption coefficient. Gas that absorbs strongly at a frequency also emits strongly there — spectral lines appear in absorption or emission depending on the temperature gradient, not on the opacity alone.
- ▶ Why it must hold (exercise 5.1): inside a blackbody enclosure at temperature T the radiation field is B and is unchanged by the presence of any body at the same T ; the transfer equation $dI/ds = -\alpha_{\text{abs}}I + \epsilon$ with $I = B$ constant then demands $\epsilon = \alpha_{\text{abs}}B$ for every frequency and direction.
- ▶ Strictly valid only in complete thermodynamic equilibrium. In planetary atmospheres we justify it for small, localized regions where the gradients of the thermodynamic properties are small compared to a photon mean free path.
- ▶ Note what Kirchhoff's law does *not* say: it does not make $I = B$. The emissivity is tied to the local temperature; the intensity is whatever the transfer equation produces.

The radiative transfer equation along a ray

Radiative transfer describes how a beam changes as it travels a distance s through a volume of gas: losses αI and gains ε . For a static atmosphere,

$$\frac{dI(\mathbf{x}, \hat{\mathbf{n}}, \nu)}{ds} = -\alpha(\mathbf{x}, \nu) I(\mathbf{x}, \hat{\mathbf{n}}, \nu) + \varepsilon(\mathbf{x}, \hat{\mathbf{n}}, \nu)$$

Time-dependent derivation. In an inertial frame, follow the energy in a fixed volume element of length ds and cross section dA during dt ; for moving material α and ε acquire an angle dependence through the Doppler shift. Balancing losses and gains,

$$[I(\mathbf{x} + \Delta\mathbf{x}, \hat{\mathbf{n}}, \nu, t + \Delta t) - I(\mathbf{x}, \hat{\mathbf{n}}, \nu, t)] dA d\Omega d\nu dt = [-\alpha I + \varepsilon] ds dA d\Omega d\nu dt,$$

and with $\Delta t = \Delta s/c$ the left side is $(\frac{1}{c}\partial_t I + \partial_s I)ds$, giving

$$\left(\frac{1}{c}\frac{\partial}{\partial t} + \frac{\partial}{\partial s}\right) I(\mathbf{x}, \hat{\mathbf{n}}, \nu, t) = -\alpha(\mathbf{x}, \hat{\mathbf{n}}, \nu, t) I + \varepsilon(\mathbf{x}, \hat{\mathbf{n}}, \nu, t).$$

- The ∂_t/c term matters only when conditions change on light-crossing times; for atmospheres we drop it (static case), together with the $\hat{\mathbf{n}}$ dependence of α , while remembering that radiation is an energy *flow* per unit time.
- In vacuum ($\alpha = \varepsilon = 0$) the equation says I is constant along the ray, as claimed last lecture.

The plane-parallel approximation

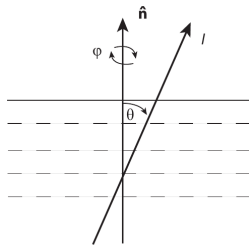


Figure 1: A 1D plane-parallel atmosphere: stratified layers, vertical coordinate z , ray at angle θ from the normal $\hat{n}$, azimuth ϕ . Reproduced from Seager (2010), Fig. 5.2.

The plane-parallel atmosphere is a good framework in which to study the radiative transfer equation.

- ▶ For a 1D planar atmosphere, the atmosphere is defined by a stratified plane with each layer having homogeneous properties such as T , P , and ρ . This 1D plane represents one location on the surface of the planet.
- ▶ **When is it valid?** If the radial depth of the atmosphere is much smaller than the planetary radius. The photosphere spans a few scale heights: $H/R_p \sim 0.01$ for hot Jupiters, 10^{-3} for Earth, so the curvature of the layers is negligible for rays that are not nearly horizontal.
- ▶ The plane-parallel definition assumes **axial symmetry** about the vertical: there is no preferred horizontal direction in the medium itself (the illumination may still come from a specific direction).
- ▶ **The vertical coordinate z increases upward; the optical depth τ increases downward from $\tau = 0$ at the top.**

The plane-parallel transfer equation

Considering the axial symmetry and adopting z as the (1D) vertical coordinate,

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial y} = 0 \quad \text{and} \quad \frac{dz}{ds} = \cos \theta \equiv \mu,$$

where θ is the angle between the surface normal and the beam of intensity. Substituting into the equation along a ray,

$$\mu \frac{dI(z, \hat{\mathbf{n}}, \nu)}{dz} = -\alpha(z, \nu) I(z, \hat{\mathbf{n}}, \nu) + \varepsilon(z, \hat{\mathbf{n}}, \nu)$$

If we further consider axial symmetry of the radiation field, i.e. no ϕ dependence,

$$\mu \frac{dI(z, \mu, \nu)}{dz} = -\alpha(z, \nu) I(z, \mu, \nu) + \varepsilon(z, \mu, \nu),$$

and a single angle variable $\mu \in [-1, 1]$ describes the direction: $\mu > 0$ upward, $\mu < 0$ downward, $\mu = 0$ horizontal.

- ▶ The factor μ is the whole geometric content of the approximation: a slant ray traverses $ds = dz/\mu$ of path for each dz of height, so it sees $1/\mu$ times more optical depth.
- ▶ Horizontal rays ($\mu \rightarrow 0$) travel infinitely far in a plane-parallel atmosphere; in reality they leave the curved atmosphere after a path $\sim \sqrt{2\pi R_p H}$, the transit chord.

The scattering integral and the limits of plane-parallel geometry

As written, the plane-parallel equation is an *integro-differential* equation, because $\varepsilon = \varepsilon_{\text{scat}} + \varepsilon_{\text{therm}}$ and the scattering term contains angle integrals of I :

$$\varepsilon_{\text{scat}}(z, \mu, \nu) = \alpha_{\text{scat}}(z, \nu) \int_0^{2\pi} \int_{-1}^1 \mathcal{P}(\mu, \phi; \mu', \phi') I(z, \mu', \nu) d\mu' d\phi'.$$

Even with azimuthal symmetry of the medium, the phase function $\mathcal{P} = P/4\pi$ couples every direction to every other direction, so the equation for $I(\mu)$ at one angle involves I at all angles.

Where the plane-parallel picture fails.

- ▶ **Transmission at the terminator:** the starlight travels along a chord, and the geometry is inherently spherical; the slant path length $\sqrt{2\pi R_p H}$ comes from the curvature.
- ▶ **Grazing incidence near the limb** of the dayside, where the stellar zenith angle approaches 90° and $1/\mu_0$ diverges; spherical (or pseudo-spherical) corrections are needed.
- ▶ **Extended atmospheres** inflated by escape or by very low gravity, where H/R_p is no longer small.
- ▶ **Horizontal inhomogeneity:** day–night contrasts, patchy clouds, and the eclipse-mapping problem require 3D or at least a collection of 1D columns.

The source function

The **source function** (sometimes called the contribution function) is the ratio of the emission coefficient to the extinction coefficient:

$$S(\mathbf{x}, \hat{\mathbf{n}}, \nu) = \frac{\varepsilon(\mathbf{x}, \hat{\mathbf{n}}, \nu)}{\alpha_{\text{ext}}(\mathbf{x}, \nu)}$$

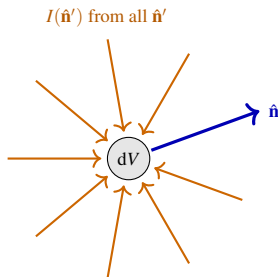
It has the same dimensions as the intensity, it simplifies the transfer equation and its solution, and it provides physical insight: it is the intensity that the beam *relaxes toward* as it travels through the medium (a beam brighter than S is attenuated, a beam fainter than S is filled in).

Expand the source function into a thermal-emission component and a scattering component. With Kirchhoff's law $\varepsilon_{\text{therm}} = \alpha_{\text{abs}} B$ and isotropic coherent scattering $\varepsilon_{\text{scat}} = \alpha_{\text{scat}} J$, under conditions of LTE,

$$S(\mathbf{x}, \nu) = \frac{\alpha_{\text{abs}}(\mathbf{x}, \nu) B(\mathbf{x}, \nu) + \alpha_{\text{scat}}(\mathbf{x}, \nu) J(\mathbf{x}, \nu)}{\alpha_{\text{ext}}(\mathbf{x}, \nu)} \iff \boxed{S = (1 - \omega_0) B + \omega_0 J}$$

with the single-scattering albedo $\omega_0 = \alpha_{\text{scat}}/\alpha_{\text{ext}}$: a weighted mean of the Planck function and the mean intensity.

Why $\epsilon_{\text{scat}} = \alpha_{\text{scat}} J$



each beam loses $\alpha_{\text{scat}} I(\hat{\mathbf{n}}')$;
the power is re-emitted
equally into all 4π sr

Energy is conserved:

$\oint \epsilon_{\text{scat}} d\Omega = 4\pi \alpha_{\text{scat}} J$ equals what was
removed. The analogy with

$\epsilon_{\text{therm}} = \alpha_{\text{abs}} B$: absorption re-emits at the
local B , scattering at the local J .

Out of every beam. A beam from direction $\hat{\mathbf{n}}'$ loses, per unit volume, time, frequency and solid angle, $\alpha_{\text{scat}} I(\hat{\mathbf{n}}')$ by scattering.
Into one beam. Coherent scattering keeps the frequency; the phase function $\mathcal{P}(\hat{\mathbf{n}}', \hat{\mathbf{n}})$ (normalized, $\oint \mathcal{P} d\Omega = 1$) says where it goes.
The total scattered power is redistributed into all directions, so the emission coefficient in direction $\hat{\mathbf{n}}$ is

$$\epsilon_{\text{scat}}(\hat{\mathbf{n}}) = \alpha_{\text{scat}} \oint I(\hat{\mathbf{n}}') \mathcal{P}(\hat{\mathbf{n}}', \hat{\mathbf{n}}) d\Omega'.$$

Isotropic scattering, $\mathcal{P} = 1/4\pi$: every direction receives the same share $1/4\pi$ of the total,

$$\epsilon_{\text{scat}} = \frac{4\pi \alpha_{\text{scat}} J}{4\pi} = \alpha_{\text{scat}} J$$

The source function: limits and consequences

- ▶ **Pure absorption** ($\omega_0 = 0$): the strict LTE value $S = B$, from Kirchhoff's law with no scattering. The beam relaxes toward the local Planck function; the emergent spectrum then reflects the temperature structure directly.
- ▶ **Pure scattering** ($\omega_0 = 1$): $S = J$, the mean intensity. The beam relaxes toward the local radiation field itself; the medium neither creates nor destroys photons, it only redistributes their directions.
- ▶ The scattering term makes S depend on the radiation field: the transfer equation becomes an *integro-differential* equation for I . Even the expression with $\omega_0 \neq 0$ “allows a slight deviation from LTE” in Seager's words: the level populations are still thermal, but the source function is not B .
- ▶ While $I = B$ for blackbody radiation, LTE radiation is *not* necessarily blackbody radiation: in LTE, I is not required to equal B . LTE constrains the *matter* (Kirchhoff, Boltzmann, Saha); the *radiation* is whatever the transfer equation produces.
- ▶ **Convention check.** Heng writes $S = \omega_0 J / 4\pi + (1 - \omega_0)B$ because his $J \equiv \oint I d\Omega$ is 4π times Seager's mean intensity $J = \frac{1}{4\pi} \oint I d\Omega$. Same physics; check which J a paper uses before comparing formulas.

Moments and the equation in optical depth

In plane-parallel geometry with azimuthal symmetry the mean intensity and flux of Chapter 2 become

$$J(z, \nu) = \frac{1}{4\pi} \int_0^{2\pi} \int_{-1}^1 I d\mu d\phi = \frac{1}{2} \int_{-1}^1 I(z, \mu, \nu) d\mu, \quad F(z, \nu) = \int_0^{2\pi} \int_{-1}^1 \mu I d\mu d\phi = 2\pi \int_{-1}^1 \mu I(z, \mu, \nu) d\mu.$$

(The stellar-atmosphere literature uses $H = F/4\pi$ because its form parallels J and K .)

Now change to the optical-depth scale ($d\tau_\nu = -\alpha dz$) and use the source function:

$$\mu \frac{dI(\tau_\nu, \mu, \nu)}{d\tau_\nu} = I(\tau_\nu, \mu, \nu) - S(\tau_\nu, \mu, \nu).$$

Note the roles of the two hemispheres: **outgoing** rays have $\mu > 0$, **incoming** rays $\mu < 0$.

Boundary conditions.

- ▶ Top ($\tau_\nu = 0$): the stellar radiation incident on the atmosphere, $I(0, \mu, \nu) = I_*(0, \mu_0, \nu)$ for $-1 \leq \mu \leq 0$, a plane-parallel beam at the single angle μ_0 . For the nightside, for infrared calculations where the planet's own emission dominates, or for unirradiated planets, $I_* = 0$.
- ▶ Bottom ($\tau_{\max, \nu}$): the intensity from the planet's interior, $I(\tau_{\max, \nu}, \mu, \nu) = I_{\text{int}}$ for $0 \leq \mu \leq 1$, matched to the interior energy flux (or, for a terrestrial planet, the surface emission and reflection).

The formal solution

Multiply the equation by the integrating factor $e^{-\tau_v/\mu}$:

$$\frac{dI}{d\tau_v} e^{-\tau_v/\mu} - \frac{I}{\mu} e^{-\tau_v/\mu} = \frac{d}{d\tau_v} \left(I e^{-\tau_v/\mu} \right) = -\frac{S}{\mu} e^{-\tau_v/\mu}.$$

Integrating from an initial optical depth $\tau_{v,i}$ to a final one $\tau_{v,f}$,

$$I(\tau_{v,f}, \mu, \nu) = I(\tau_{v,i}, \mu, \nu) e^{-(\tau_{v,i} - \tau_{v,f})/\mu} - \frac{1}{\mu} \int_{\tau_{v,i}}^{\tau_{v,f}} S(\tau_v, \mu, \nu) e^{-(\tau_v - \tau_{v,f})/\mu} d\tau_v$$

- **First term:** the initial intensity diminished by the exponential attenuation of absorption over the optical path $(\tau_i - \tau_f)/\mu$.
- **Second term:** the emission from the atmosphere, an exponentially weighted average of the source function along the beam up to the location of interest.
- For an outgoing ray ($\mu > 0$) from deep inside ($\tau_i > \tau_f$) the integral runs downward in τ and the second term is positive: each layer contributes $S d\tau/\mu$, attenuated by $e^{-(\tau - \tau_f)/\mu}$ on its way up. For an incoming ray ($\mu < 0$) the same formula, read from the top down, describes the penetration of starlight with the atmosphere's emission added.

It is called the “formal” solution because S must be known; in general it is not.

The emergent intensity

Our goal is the planet's spectrum: the emergent flux at the top of the atmosphere, the measurable quantity. For a *semi-infinite* atmosphere, integrate from deep inside, $\tau_{v,i} = \infty$, to the top, $\tau_{v,f} = 0$, using $\lim_{\tau_v \rightarrow \infty} I e^{-\tau_v/\mu} = 0$ (the deep layers are hidden):

$$I(0, \mu, \nu) = \frac{1}{\mu} \int_0^\infty S(\tau_v, \mu, \nu) e^{-\tau_v/\mu} d\tau_v \quad (0 < \mu \leq 1).$$

The emergent intensity is the amount of intensity from each altitude that reaches the surface along a path at angle θ to the line of sight.

- The weight $e^{-\tau/\mu}/\mu$ is normalized ($\int_0^\infty e^{-\tau/\mu} d\tau/\mu = 1$) and peaks at the top, with mean $\langle \tau_v \rangle = \mu$: the emergent intensity is a weighted sample of S over $\tau_v \lesssim \mu$, the “photosphere” at that frequency and angle. Rays emerging at grazing angles ($\mu \rightarrow 0$) sample shallower, usually cooler layers — this is **limb darkening**.

Why the problem is hard

- **The source function depends on I through scattering.** $S = (1 - \omega_0)B + \omega_0 J$, or with the angle-dependent scattering emissivity

$$\varepsilon_{\text{scat}}(\tau, \mu, \nu) = \sigma_s(\tau, \nu) \int_0^{2\pi} \int_{-1}^1 \mathcal{P}(\mu, \phi; \mu', \phi') I(\tau, \mu', \nu) d\mu' d\phi',$$

and J is an integral of the unknown I . Physically, scattering *decouples* the intensity from the local conditions: photons may scatter through large distances in the atmosphere without interacting with the thermal pool of the gas via absorption and thermal reemission. A numerical solution of an integro-differential equation is required.

- **Even B depends on I** through the temperature structure: radiative equilibrium ties $T(\tau)$ to the absorbed radiation (next section), so S is unknown until the whole problem, including the energy balance of every layer, is solved.

[Temperature, Albedos, and Flux Ratios] - Chapter 3

When observing a distant exoplanet, the only thing we can measure is the radiation coming from it: a *flux*. The quantity relevant for the physics and chemistry of the atmosphere is its *temperature*.

- ▶ Only a single flux may be measured, yet real planets rarely have a single temperature. Which temperature should represent the planet? Which temperature should represent the flux?
- ▶ Temperature and albedo are tied together by **energy balance**: what is not reflected is absorbed and must be reradiated.
- ▶ Planet-to-star flux ratios (next lecture) follow from the temperatures and albedos, and decide what kind and size of telescope is needed to detect a given planet.

This section: energy balance $\rightarrow$ three planetary temperatures (T_{eff} , T_{eq} , T_{b}) $\rightarrow$ four albedos (single-scattering, geometric, spherical, Bond) and the relations among them.

Energy balance

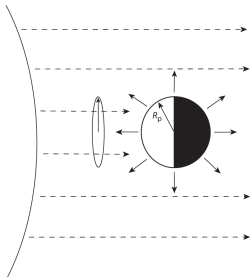


Figure 2: Planetary energy balance. Stellar radiation falls on the planet's cross-section πR_p^2 ; a fraction $(1 - A_B)$ is absorbed, here advected around the planet and reemitted uniformly into 4π . Reproduced from Seager (2010), Fig. 3.1.

No energy is created or destroyed in the atmosphere. All the energy comes either from the star (absorbed incident radiation) or from the planetary interior:

$$E_{\text{out}}(t) = (1 - A_B) E_{\text{inc}}(t) + E_{\text{int}}(t)$$

- ▶ E_{out} : energy per unit time leaving the planet.
- ▶ E_{inc} : stellar energy per unit time incident on the planet.
- ▶ $(1 - A_B)$: fraction absorbed, heating the atmosphere or surface, later reemitted at longer wavelengths.
- ▶ A_B : the **Bond albedo**, the fraction of incident stellar energy scattered back into space.
- ▶ E_{int} : energy per unit time supplied by the interior.

Energy balance in terms of fluxes

Write the outgoing energy as the planetary luminosity $L_p = 4\pi R_p^2 F_{S,p}$ and the incident energy with the result of Chapter 2, $E_{\text{inc}} = \pi R_p^2 (R_*/a)^2 F_{S,*}$:

$$4\pi R_p^2 F_{S,p} = (1 - A_B) F_{S,*} \left(\frac{R_*}{a} \right)^2 \pi R_p^2 + L_{p,\text{int}}$$

Two energy sources for the atmosphere: (1) absorbed (and reradiated) starlight, and (2) the planet's own interior luminosity $L_{p,\text{int}}$, the flux passing through a surface enclosing the planet from the inside.

- **Giant planets** (H/He): $L_{p,\text{int}}$ is the gradual loss of residual gravitational potential energy from the planet's formation (Kelvin–Helmholtz contraction, plus helium rain in Saturn). **Jupiter's internal luminosity is more than *twice* its reradiated absorbed sunlight** — an indication of how long it takes energy to travel from a giant planet's interior out to space.
- **Earth**: $L_{p,\text{int}}$ ($\simeq 44$ TW) comes partly from residual accretion energy but mostly from the decay of radioactive isotopes of uranium (U, 92), thorium (Th, 90), and potassium (K, 19). It is $\sim 2 \times 10^{-4}$ of the absorbed sunlight (1.2×10^{17} W, exercise 3.1b): negligible for the surface temperature, but it drives plate tectonics and volcanism, and thereby the long-term atmospheric evolution.

When can the interior be neglected?

For many planets the luminosity from external heating overwhelms the interior luminosity by many orders of magnitude:

- ▶ **Hot Jupiters** in very short-period orbits ($P < 4$ d): with $T_{\text{eq}} = 1600$ K and a Jupiter-like interior heat flow ($T_{\text{eff,int}} \simeq 120$ K), the ratio of interior to absorbed stellar energy is $(120/1600)^4 \simeq 3 \times 10^{-5}$ (exercise 3.1a).
- ▶ **Terrestrial planets**, with their typically limited interior energy (Earth: 2×10^{-4}).
- ▶ **Exceptions:** young giant planets and brown dwarfs (directly imaged planets are seen *because* of their interior luminosity), and the Solar System giants, where the interior contributes 30–100% of the emitted power.

We therefore usually write the energy balance without the interior term,

$$4\pi R_p^2 F_{S,p} = (1 - A_B) F_{S,*} \left(\frac{R_*}{a} \right)^2 \pi R_p^2$$

Time dependence and eccentric orbits

- ▶ The planet exists in an equilibrium with the incident stellar radiation in which the heating is constant in time, so from now on we drop the time dependence of the planet flux and related quantities. (Energy per unit time is technically called *power*.)
- ▶ Planets on **eccentric orbits** do *not* have constant heating: the incident flux varies as $1/r^2$ between periastron and apastron, by a factor $[(1+e)/(1-e)]^2$, which is 2.3 for $e = 0.2$ and 1000 for HD 80606b ($e = 0.93$).
- ▶ Exercise 3.3: replacing a by the instantaneous distance r and averaging the incident flux over the orbit gives the orbit-averaged equilibrium temperature $\langle T_{\text{eq}} \rangle = T_{\text{eq}}(a) (1 - e^2)^{-1/8}$, a weak correction because the planet spends most of its time near apastron.
- ▶ Whether the atmosphere follows the instantaneous heating or the average depends on the **radiative timescale** of the photosphere compared with the orbital period: hot, low-pressure photospheres respond within hours (HD 80606b's $8\text{ }\mu\text{m}$ flux rises by a factor of 5 within 6 hours of periastron passage), whereas deep, cold layers average over many orbits.

Three temperatures for a planet

There is no single temperature that describes a planet atmosphere: it varies with altitude, with horizontal position, and in time (day/night, seasons). Yet temperature governs, to first order, the chemical equilibrium state and hence the emergent spectrum. Three temperatures are in common use:

- ▶ **Effective temperature** T_{eff} — a proxy for the global temperature; defined from the *total* emitted flux. In principle a measured quantity.
- ▶ **Equilibrium temperature** T_{eq} — the T_{eff} a planet with no internal luminosity would have in equilibrium with its star. A *theoretical* number computed from R_* , $T_{\text{eff},*}$, a , and A_B .
- ▶ **Brightness temperature** $T_b(\nu)$ — the temperature of a blackbody with the same flux *at a given frequency*. The only temperature we can currently *measure* for exoplanets.

All three are defined by comparing the planet to a blackbody of the same size and distance; they differ in *which* flux is matched (total or monochromatic, measured or theoretical).

The effective temperature T_{eff}

T_{eff} is the temperature of a blackbody at the same distance as the planet with the *same total flux* (integrated over all frequencies). Using the blackbody flux πB and assuming a uniform surface flux,

$$F_S \equiv \int_0^\infty F_S(\nu) d\nu = \pi \int_0^\infty B(T, \nu) d\nu \equiv \sigma_R T_{\text{eff}}^4.$$

Carrying out the integral (exercise 3.2, using $\int_0^\infty u^3/(e^u - 1) du = \pi^4/15$) gives the **Stefan–Boltzmann law** and constant

$$F_S = \sigma_R T_{\text{eff}}^4, \quad \sigma_R = \frac{2\pi^5 k_B^4}{15c^2 h^3} = 5.670 \times 10^{-8} \text{ J K}^{-4} \text{ m}^{-2} \text{ s}^{-1}$$

- ▶ σ_R is not a fundamental constant: it is built from h , c , and k_B . (Heng, problem 2.8.4: the same $\sigma_R T^4$ results whether one integrates B_λ , B_ν , or $B_{\tilde{\nu}}$.)
- ▶ In principle T_{eff} is a *measured* quantity: take the total flux of the planet, convert it to surface flux with $(D_\oplus/R_p)^2$, and find the temperature of the blackbody with the same total flux. In practice we never observe *all* wavelengths, so a model is needed to fill the gaps.
- ▶ F_S here is the planetary surface flux, in $\text{J m}^{-2} \text{ s}^{-1}$, radiated at the planet, as derived in Chapter 2.

What T_{eff} means physically

- ▶ T_{eff} describes the global temperature of the planet at the altitude where the *bulk of the radiation leaves* the planet or star. This altitude is sometimes called the planet's **photosphere** or “surface”, whether or not it is at the solid surface.
- ▶ T_{eff} may be very different from the surface temperature. **Venus:** $T_{\text{eff}} \simeq 230$ K, a value that comes from the cloud tops which obscure the solid surface. Because of a strong greenhouse effect the surface temperature is 730 K, almost 500 K hotter than T_{eff} . **Earth:** $T_{\text{eff}} \simeq 255$ K versus a mean surface temperature of 288 K, a greenhouse warming of 33 K.
- ▶ T_{eff} varies widely among planets. The coldest planets of the Solar System, Uranus and Neptune, have $T_{\text{eff}} = 59$ K; Jupiter 124 K; Earth $\simeq 255$ K. If we could measure T_{eff} for exoplanets, we would expect the hottest hot Jupiters to have $T_{\text{eff}} > 2000$ K, and ultra-hot Jupiters such as KELT-9b exceed 4000 K on the dayside.
- ▶ Because T_{eff} is defined by the *total* flux, it says nothing about the spectral shape (next slide).

T_{eff} applied to two stars

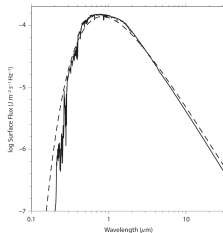


Figure 3: Surface flux of a $T_{\text{eff}} = 6130$ K G0IV star (HD 149026; solid) and the 6130 K blackbody (dashed). Seager (2010), Fig. 3.2.

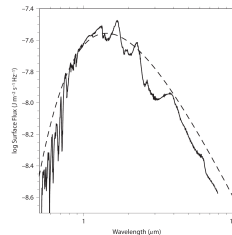


Figure 4: Same for a $T_{\text{eff}} = 3600$ K M2.5V star (GJ 436); the deep features are TiO (visible) and H₂O (near-IR). Seager (2010), Fig. 3.3.

Both stars have the same total flux as their blackbodies (equal areas under the curves). For the G star the spectrum is close to a blackbody; for the M star it departs strongly. *Lesson:* T_{eff} says nothing about the spectral shape. A planet around GJ 436 receives far more near-infrared and far less blue light than a blackbody would suggest, which matters for the Bond albedo (weighted by the stellar spectrum), for the photochemistry (little UV), and for the interpretation of secondary-eclipse depths (the stellar brightness temperature in the bands is not T_{eff}).

The equilibrium temperature T_{eq}

T_{eq} is an estimate of T_{eff} for a planet with *no internal luminosity*: the effective temperature attained by an isothermal planet in complete equilibrium with the radiation of its star. It refers to no flux measurement and is essential for exoplanets, for which T_{eff} is hard to measure. Start from energy balance, with a correction factor f to the 4π so that we can describe one hemisphere:

$$\frac{4\pi}{f} R_p^2 F_{S,p} = (1 - A_B) F_{S,*} \left(\frac{R_*}{a} \right)^2 \pi R_p^2.$$

Insert the Stefan–Boltzmann law $F_{S,p} = \sigma_R T_{\text{eq}}^4$ and $F_{S,*} = \sigma_R T_{\text{eff},*}^4$ ($T_{\text{eff},*}$ = effective temperature of the host star) and define $f' = f/4$:

$$T_{\text{eq}} = T_{\text{eff},*} \left(\frac{R_*}{a} \right)^{1/2} [f'(1 - A_B)]^{1/4}$$

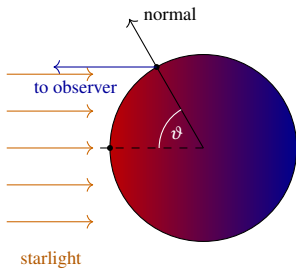
- **Scaling:** $T_{\text{eq}} \propto a^{-1/2}$ and $T_{\text{eq}} \propto R_*^{1/2} T_{\text{eff},*}$; the albedo enters only as $(1 - A_B)^{1/4}$ (a factor 0.9 for $A_B = 0.3$).
- Since $L_* = 4\pi R_*^2 \sigma_R T_{\text{eff},*}^4$, equivalently $T_{\text{eq}}^4 = f'(1 - A_B) L_* / (4\pi \sigma_R a^2)$: the temperature is set by the stellar flux at the planet's orbit.

The redistribution factor f'

For exoplanets we observe only one hemisphere at a time; this is what drives the need for f' . It describes *how the absorbed stellar energy is distributed* over the planet before being reradiated.

- ▶ **Uniform redistribution** ($f' = 1/4, f = 1$): the absorbed energy is spread over the whole sphere ($4\pi R_p^2$) and T_{eq} is the same for any hemispheric view. For Earth: $T_{\text{eq}} = 5778 \text{ K } (R_{\odot}/\text{AU})^{1/2} [0.7/4]^{1/4} = 255 \text{ K}$.
- ▶ **Instantaneous reradiation, no advection** ($f' = 2/3, f = 8/3$): a slowly rotating or tidally locked planet absorbs energy only on the dayside and each element reradiates locally. The dayside hemisphere is then much hotter: $(8/3)^{1/4} = 1.28$ times the uniform value. The value $2/3$ is the same number as the geometric albedo of a Lambert sphere, and arises from the same $\cos \vartheta$ -weighted hemispheric integral: the observer sees the hot substellar region face-on and the cooler limb obliquely.
- ▶ Between these two end cases there is no simple analytic form. To estimate the actual dayside temperature one needs observations (phase curves) or atmospheric circulation models (Chapter 10).
- ▶ **Caution:** f has many different definitions in the literature ($f = 1/4, 1/2, 2/3, \dots$ meaning different things). Seager's choice $f' = f/4$ as a correction to 4π is deliberate, for symmetry with the albedo derivation for scattered light from the dayside.

Where $f = 8/3$ comes from: two factors of $\cos \vartheta$



Dayside of a tidally locked planet at secondary eclipse: the substellar point faces both the star and the observer, the limb is oblique to both.

1. Local energy balance. With no advection, each surface element radiates exactly what it absorbs. An element whose normal makes the angle ϑ with the direction to the star intercepts the stellar flux foreshortened by $\cos \vartheta$:

$$\sigma T^4(\vartheta) = (1 - A_B) F_{\text{inc}} \cos \vartheta \quad \implies \quad T(\vartheta) = T_0 \cos^{1/4} \vartheta,$$

with $T_0^4 = (1 - A_B) F_{\text{inc}} / \sigma$; T_0 is the substellar temperature. The dayside is therefore not isothermal: it is hottest where the star stands overhead and falls to zero at the terminator, where the starlight arrives grazing.

Where $f = 8/3$ comes from: the two integrals

2. What the observer collects. The emitted flux of each element is foreshortened once more, by $\cos \vartheta$ as seen from Earth, and the dayside is normalized by its projected area πR_p^2 :

$$T_{\text{day}}^4 = \frac{1}{\int_0^{\pi/2} \cos \vartheta \sin \vartheta d\vartheta} \int_0^{\pi/2} T^4(\vartheta) \cos \vartheta \sin \vartheta d\vartheta = 2 T_0^4 \int_0^{\pi/2} \cos^2 \vartheta \sin \vartheta d\vartheta = \frac{2}{3} T_0^4.$$

Here, we note that $\int_0^{\pi/2} \cos \vartheta \sin \vartheta d\vartheta = 1/2$ and $\int_0^{\pi/2} \cos^2 \vartheta \sin \vartheta d\vartheta = 1/3$.

3. Comparison with the uniform case. Uniform redistribution spreads the same absorbed power over the whole sphere, i.e. over four times the intercepting area, so $\sigma T_{\text{unif}}^4 = (1 - A_B) F_{\text{inc}}/4$ and $T_0^4 = 4 T_{\text{unif}}^4$. Substituting this into $T_{\text{day}}^4 = \frac{2}{3} T_0^4$,

$$T_{\text{day}}^4 = \frac{2}{3} \cdot 4 T_{\text{unif}}^4 = \frac{8}{3} T_{\text{unif}}^4 \quad \implies \quad f = \frac{8}{3}, \quad f' = \frac{f}{4} = \frac{2}{3} \quad \text{for instantaneous reradiation.}$$

- ▶ The factor 4 is geometric (sphere versus disk); the factor $2/3$ is the $\cos \vartheta$ -weighted average of the $\cos \vartheta$ heating pattern over the visible hemisphere.
- ▶ In temperature, $(8/3)^{1/4} = 1.28$: the dayside of a planet with no redistribution is 28% hotter than the same planet with a uniform surface temperature.
- ▶ The same double- $\cos \vartheta$ integral gives $A_g = 2/3$ for a Lambert sphere, which is why the two numbers coincide: an isotropically emitting (or scattering) surface illuminated as $\cos \vartheta$ and viewed at $\cos \vartheta$.
- ▶ **Real planets lie between $f = 1$ and $f = 8/3$** ; winds move heat to the nightside and lower the dayside temperature. Measuring where a planet falls between the two is exactly what a secondary eclipse plus a phase curve delivers.

T_{eq} versus orbital distance

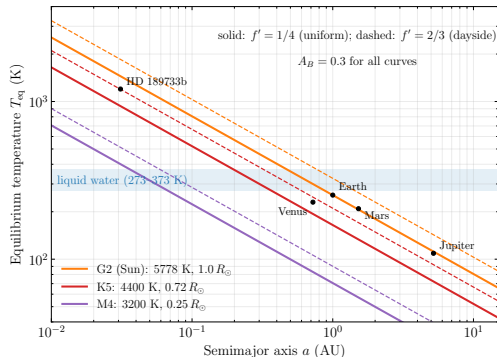


Figure 5: Equilibrium temperature versus semimajor axis for three host stars, with $A_B = 0.3$; solid $f' = 1/4$, dashed $f' = 2/3$. Points are Solar System T_{eq} values (Table 3.1) and HD 189733b.

- ▶ The $a^{-1/2}$ slope: moving a planet 100 times closer raises T_{eq} by a factor of 10 (Earth at 0.01 AU would be ~ 2500 K).
- ▶ Around an M4 dwarf the same T_{eq} is reached at $a \simeq 0.05$ AU: the habitable zone shrinks inward together with T_{eq} , which is why M-dwarf habitable-zone planets transit frequently and are tidally locked.
- ▶ HD 189733b ($a = 0.031$ AU, K dwarf): $T_{\text{eq}} \simeq 1200$ K for $f' = 1/4$, close to the measured dayside brightness temperature (Lecture 1), indicating efficient day–night heat transport.
- ▶ The dashed curves show how much hotter a non-redistributing dayside can be.
- ▶ The blue band marks 273–373 K, where water is liquid at 1 bar: a crude habitable-zone gauge, since T_{eq} is not the surface temperature.^a

^aGreenhouse warming lifts the surface above T_{eq} : Earth $255 \rightarrow 288$ K, Venus $230 \rightarrow 735$ K.

How good is T_{eq} ? The Solar System test

Planet	T_{eff} (K)	T_{eq} (K)	A_g (500 nm)	A_B
Mercury	—	—	0.106	0.119
Venus	~ 230	230	0.65	0.750
Earth	~ 255	253	0.367	0.306
Mars	~ 212	209	0.150	0.250
Jupiter	124.4 ± 0.3	109	0.52	0.343
Saturn	95.0 ± 0.4	80	0.47	0.342
Uranus	59.1 ± 0.3	58	0.51	0.300
Neptune	59.3 ± 0.8	46	0.42	0.290

Seager (2010), Table 3.1; values from Cox (2000). T_{eq} uses the measured A_B , the known a , and $f' = 1/4$.

- For the terrestrial planets T_{eff} and T_{eq} agree to within a few kelvin: their interior heat is negligible.
- For Jupiter, Saturn, and Neptune $T_{\text{eff}} > T_{\text{eq}}$ by 15–30%, the signature of their **internal luminosity** ($T_{\text{eff}}^4 = T_{\text{eq}}^4 + T_{\text{int}}^4$ with $T_{\text{int}} \simeq 100$ K for Jupiter). Uranus is the exception: $T_{\text{eff}} \simeq T_{\text{eq}}$, so its interior heat flow is anomalously small — still not fully understood.
- Note that $A_g \neq A_B$ in general, and that A_g can exceed A_B (Jupiter, Uranus, Neptune): the two albedos measure different things (next section).

The brightness temperature $T_b(\nu)$: motivation

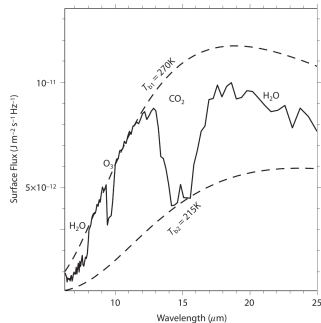


Figure 6: Brightness temperature. Solid: Earth's surface flux measured by Mars Global Surveyor. Dashed: blackbody fluxes at two temperatures. Reproduced from Seager (2010), Fig. 3.4.

The flux of exoplanets at different wavelengths or frequencies is the quantity we can measure at Earth, $F_{\oplus} = (R_p/D_{\oplus})^2 F_S$ for a very distant planet with surface flux F_S .

- Our intuitive familiarity with temperature makes it easier to use a temperature instead of a flux to describe exoplanet atmosphere measurements.
- Temperatures are more directly relevant than fluxes for the physics and chemistry of the atmosphere.
- Fluxes depend on the planet's distance from Earth and so change from one planet to the next, whereas **the temperature of a planet atmosphere is distance independent.**

Measured exoplanet fluxes are therefore often stated as **brightness temperatures**: the temperature of a blackbody of the same shape and at the same distance as the planet with the *same flux in a specified frequency range*.

The brightness temperature $T_b(\nu)$: definition

$$\pi B(T_b(\nu), \nu) = F_S(\nu) = F_{\oplus}(\nu) \left(\frac{D_{\oplus}}{R_p} \right)^2$$

- ▶ $T_b(\nu)$ is a frequency-dependent expression of the planetary flux: invert the Planck function at the observed frequency for the temperature. Seager writes this with the Stefan–Boltzmann form $\sigma_R T_b^4(\nu) = F_S(\nu)$ as shorthand for “convert flux to temperature”; the monochromatic Planck inversion is the operational definition used in practice.
- ▶ The brightness temperature assumes that the planet behaves as a blackbody *only* in the frequency range where the flux is measured; nothing is assumed about other frequencies.
- ▶ While a blackbody radiator has a constant $T_b(\nu)$, planets with spectra that depart from a blackbody have a $T_b(\nu)$ that can vary widely with wavelength or frequency.
- ▶ In the Rayleigh–Jeans regime T_b is simply proportional to the flux, $T_b = c^2 F_S(\nu) / (2\pi \nu^2 k_B)$, which is the definition used in radio astronomy; in the Wien regime a small change in flux corresponds to a small change in T_b , so brightness temperatures are a compressed, robust way to quote infrared fluxes.

Reading brightness temperatures

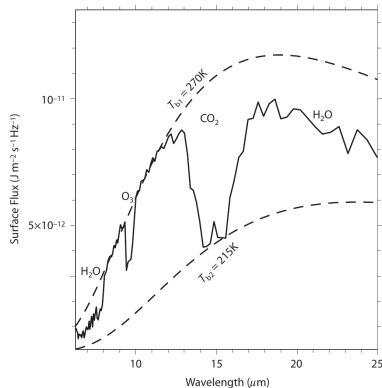


Figure 7: Earth's mid-infrared spectrum:
 $T_b(12\,\mu\text{m}) \simeq 270\,\text{K}$ but $T_b(14.2\,\mu\text{m}) \simeq 215\,\text{K}$.
Reproduced from Seager (2010), Fig. 3.4.

- ▶ For Earth: $T_b \simeq 270\,\text{K}$ in the $12\,\mu\text{m}$ window, where we see the warm surface through a transparent atmosphere, but $T_b \simeq 215\,\text{K}$ in the $14\text{--}15\,\mu\text{m}$ CO_2 band, where the atmosphere is opaque and we see the cold tropopause (대류권계면) region. The $9.6\,\mu\text{m}$ O_3 band gives an intermediate value from the stratosphere (성층권).
- ▶ $T_b(\nu)$ is the *only* temperature we can currently measure for exoplanets, because it requires flux at just one frequency or wavelength range: a secondary-eclipse depth in the Spitzer 3.6, 4.5, 5.8, and $8\,\mu\text{m}$ bands gives four brightness temperatures of the dayside.
- ▶ To get from a measured $T_b(\nu)$ to the more global T_{eff} we must resort to a model atmosphere code that fills in the unobserved wavelengths.

Relating the three temperatures

	definition	measured or theoretical?	typical use
T_{eff}	blackbody with the same <i>total</i> flux: $\sigma_{\text{R}} T_{\text{eff}}^4 = F_{\text{S}}$	measurable in principle (needs all λ)	global energy output; interior heat
T_{eq}	T_{eff} of a planet with no interior heat, from R_* , $T_{\text{eff},*}$, a , A_{B} , f'	theoretical	first estimate for any new planet
$T_{\text{b}}(\nu)$	blackbody with the same flux <i>at ν</i>	measured (one band suffices)	reporting eclipse depths; vertical structure via $T_{\text{b}}(\nu)$

- ▶ $T_{\text{eff}}^4 = T_{\text{eq}}^4 + T_{\text{int}}^4$ when interior heat matters (the two fluxes add); $T_{\text{eff}} \simeq T_{\text{eq}}$ otherwise, provided the right f' is used for the hemisphere observed.
- ▶ All three compare the planet with a blackbody of the same size and distance; they differ only in *which* flux is matched: total or monochromatic, measured or computed.

Temperatures of a tidally locked planet

- ▶ $T_b(\nu)$ scatters around T_{eff} : it is higher in spectral windows that see deep, warm layers and lower in strong absorption bands that see high, cold layers, for a temperature decreasing with altitude. A temperature *inversion* reverses this: the bands appear in *emission* and T_b is higher inside them. Thus $T_b(\nu)$ across a spectrum is a crude thermometer of the vertical structure, which Chapter 9 turns into a quantitative retrieval.
- ▶ For a tidally locked hot Jupiter the *dayside* T_b measured at secondary eclipse can exceed $T_{\text{eq}}(f' = 1/4)$ by up to $(8/3)^{1/4} = 1.28$ if the absorbed energy is not redistributed, whereas the *nightside* T_b measured near transit can be far below it. The day–night contrast, i.e. the effective f' , is a diagnostic of the winds (Chapter 10).
- ▶ Example: HD 189733b has $T_{\text{eq}}(f' = 1/4) \simeq 1200$ K; the measured $8\text{ }\mu\text{m}$ brightness temperatures are $\simeq 1210$ K on the dayside and $\simeq 970$ K on the nightside (Knutson et al. 2007), so heat redistribution is efficient. For WASP-18b, in contrast, the dayside is close to the $f' = 2/3$ limit and the nightside is very cold.
- ▶ A single measured T_b therefore does not determine A_B : with f' unknown, albedo and redistribution are degenerate in the dayside temperature. Breaking the degeneracy needs a phase curve or an optical eclipse depth.

What makes a planet bright or dark?

The **albedo** is the ratio of the light scattered by a planet to the light received by it. As for temperature, there is no single albedo; but **the albedo is an important parameter because the reflectivity is indicative of cloud or surface conditions, and because it controls the energy balance and hence T_{eff} .**

- ▶ **Venus** is the brightest planet, scattering 0.75 of the incident energy: 100% coverage by H_2SO_4 (sulfuric acid, 황산) clouds.
- ▶ **Earth** has highly reflective water/ice clouds but only $\sim 50\%$ cover, and scatters 0.3.
- ▶ **Icy bodies** are bright, fresh ice more than old ice. Enceladus (a small, 500 km icy moon of Saturn whose south-polar geysers vent water from a subsurface ocean and continually coat the surface with fresh snow) is extremely bright.
- ▶ **Mercury**, with no atmosphere, clouds, or ice, scatters only 0.1. Hot Jupiters are similarly dark ($A_g \lesssim 0.1$): no bright cloud decks, and strong Na/K absorption.

Four albedos and their relations: **single-scattering** $\tilde{\omega}$ (one particle), **geometric** $A_g(\nu)$ (full phase, relative to a Lambert disk), **spherical** $A_S(\nu)$ (all directions, one frequency), **Bond** A_B (all directions, all frequencies) — plus the **apparent** albedo, the one that can be measured for exoplanets.

The single-scattering albedo $\tilde{\omega}$

- ▶ The single-scattering albedo $\tilde{\omega}$ is the fraction of incident light that is scattered (rather than absorbed) by a *single particle* in the atmosphere or on the surface. It is wavelength dependent.
- ▶ Examples: a water-ice crystal in Earth's atmosphere can reach $\tilde{\omega} \simeq 0.8$; a plant leaf at visible wavelengths is closer to 0.1. Old to fresh snow 0.45–0.85; clouds 0.35–0.8; desert sand 0.2–0.35; ocean water < 0.05 .
- ▶ $\tilde{\omega}$ is *not* a global albedo of the planet. Nevertheless, the reflective properties of the individual gas, cloud, and surface particles ultimately produce the planet's global albedo through complex *multiple* scattering of the incident radiation.
- ▶ Equally important is the *directional* scattering property of a particle. Few particles in nature, if any, scatter isotropically; ice crystals, for example, have a strong tendency to backscatter. For a surface this is described by the bidirectional reflectance distribution function (BRDF); for a single particle by the phase function $P(\Theta)$ (Chapter 5).
- ▶ In radiative-transfer language, $\tilde{\omega} = \sigma_s / (\sigma_a + \sigma_s)$, the ratio of the scattering coefficient to the total extinction. Conservative scattering, $\tilde{\omega} = 1$, is the Lambert-sphere idealization; a Rayleigh-scattering ($\sigma_{\text{Ray}} \propto \lambda^{-4}$) atmosphere over a black surface has $\tilde{\omega} \rightarrow 1$ in the blue and $\rightarrow 0$ in the red.

The phase angle α

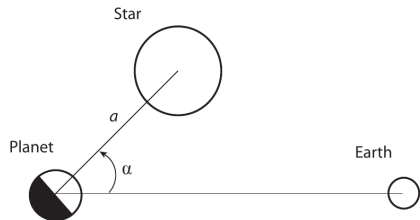


Figure 8: The phase angle α is the star–planet–observer angle; a is the semimajor axis. Reproduced from Seager (2010), Fig. 3.5.

Before defining the planetary albedos we need the **phase angle**: the star–planet–observer angle.

- ▶ $\alpha = 0^\circ$ corresponds to full phase: the observer sees the entire illuminated hemisphere.
- ▶ $\alpha = 90^\circ$ is quadrature (“half” phase); this is where the planet’s angular separation from the star is largest for a circular orbit, the natural configuration for direct imaging.
- ▶ $\alpha > 170^\circ$ corresponds to a thin crescent, and at $\alpha = 180^\circ$ the planet is not illuminated at all as seen by the observer (it is between the observer and the star: transit).

Phase angles in the Solar System and for exoplanets

- ▶ **Solar System:** only Mercury and Venus, interior to Earth's orbit, show all illumination phases from Earth; the outer giant planets show only a few to several degrees of phase angle. Full phase, $\alpha \simeq 0$, is the natural configuration for observing Solar System planets, with the Sun, Earth, and planet aligned at opposition.
- ▶ For exoplanets the range of visible planetary phases depends on the **orbital inclination** of the planet–star system.
- ▶ A planet in a system with $i = 90^\circ$ (edge-on) shows all phases as seen from Earth, except phases right near $\alpha = 0$ when the planet is directly behind the star (secondary eclipse) and near $\alpha = 180^\circ$ when it is in front (transit).
- ▶ In contrast, a planet with a face-on orbit ($i = 0^\circ$) has a constant “half” phase, $\alpha = 90^\circ$, throughout its orbit. In general the accessible range is $90^\circ - i \leq \alpha \leq 90^\circ + i$.

Full phase is never quite observable

- ▶ Because of the randomness of exoplanet orbital inclinations, $\alpha \approx 0$ may never occur for some planets.
- ▶ In reality $\alpha \equiv 0$ does not occur for any exoplanet, because the planet would be directly behind its star and not observable; however, we consider a small α close enough to $\alpha = 0$ for our purposes — for a transiting planet, the moments just before and after secondary eclipse, where $\alpha \simeq R_*/a \sim 5^\circ$.
- ▶ **Phase (φ) vs. Phase angle (α):** The phase angle is related to the orbital phase: for a circular edge-on orbit, $\cos \alpha = -\cos(2\pi\varphi)$ with φ the orbital phase measured from transit, so that α sweeps from 180° at transit through 90° at quadrature to 0 at eclipse. For inclination i , $\cos \alpha = -\sin i \cos(2\pi\varphi)$.
- ▶ For directly imaged planets the situation is reversed: small α (planet behind or in front of the star) is inaccessible because of the angular proximity to the star, and the accessible phases cluster around quadrature. Reflected-light imaging therefore measures the apparent albedo near $\alpha \simeq 90^\circ$ rather than A_g .

The geometric albedo $A_g(\nu)$: definition

The geometric albedo is the ratio of a planet's flux at *zero phase angle* (full phase) to the flux from a **Lambert disk** at the same distance and with the same cross-sectional area (i.e. subtending the same solid angle).

- ▶ The definition is historical: the brightness of a Solar System planet on a photograph was compared to that of a uniformly illuminated Lambert disk, a *relative* measurement that cancels instrumental errors.
- ▶ $A_g(\nu)$ is *not* the fraction of incident energy scattered toward $\alpha = 0$ (exercise 3.5: the fraction, per steradian, is A_g/π).

$$A_g(\nu) \equiv \frac{F_{\oplus}(\nu)}{F_{\text{L disk}}(\nu)}$$

The geometric albedo: numerator and denominator

Numerator: the planet flux at Earth at $\alpha = 0$, from Chapter 2 in latitude–longitude coordinates, keeping only *scattered* intensity (not thermal reemission):

$$F_{\oplus}(\nu) = \left(\frac{R_p}{D_{\oplus}} \right)^2 \int_{-\pi/2}^{\pi/2} \int_{-\pi/2}^{\pi/2} I_{S,\text{scat}}(\theta, \phi, \nu) \cos \phi \cos^2 \theta \, d\theta \, d\phi.$$

Denominator: a Lambert disk of radius R_p scatters isotropically with $I_{S,\text{scat}} = I_{\text{inc}}$, uniform over the disk, so

$$F_{\text{L disk}}(\nu) = \left(\frac{R_p}{D_{\oplus}} \right)^2 \pi I_{\text{inc}}(\nu).$$

Here

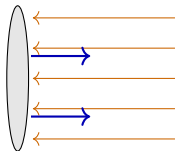
$$I_{\text{inc}}(\nu) \equiv \frac{F_{\text{inc}}(\nu)}{\pi}$$

is the intensity that a Lambert surface facing the star would have — not the intensity of the stellar beam itself, which arrives from a single direction.

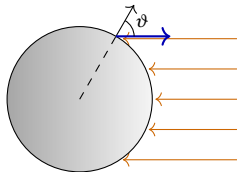
Lambert disk versus Lambert sphere

star and observer ($\alpha = 0$) $\longrightarrow$

Lambert disk



Lambert sphere



Both scatter all incident light isotropically (Lambert's law: equally bright from every viewing angle); they differ only in shape.

Lambert disk — a flat disk facing the star and the observer. Every point is lit and seen along its normal, so the disk is uniformly bright. It is the *yardstick* in the definition of the geometric albedo: $A_g \equiv 1$.

Lambert sphere — the same surface on a sphere of the same cross section. Away from the substellar point an element is lit obliquely ($\cos \vartheta$) and seen obliquely ($\cos \vartheta$), so the limb is darker than the center:

$$A_g = \frac{2}{3}, \quad q = \frac{3}{2}, \quad A_S = A_g q = 1.$$

Nothing is absorbed ($A_S = 1$); the missing third at full phase is light the curved surface sends *away from* the line of sight, which the phase integral q recovers at other phase angles.

A perfect Lambert surface has $p = 1$: energy conservation

Consider one surface element whose normal makes the angle Θ with the direction to the star, and use Seager's description $I_{S,\text{scat}} = I_{\text{inc}}(\theta', \phi') p = p \cos \Theta I_{\text{inc}}(\nu)$.

1. **Flux in.** The stellar flux through the element is foreshortened by $\cos \Theta$; with $I_{\text{inc}} \equiv F_{\text{inc}}/\pi$,

$$F_{\text{in}} = \cos \Theta F_{\text{inc}} = \pi \cos \Theta I_{\text{inc}}(\nu).$$

2. **Flux out.** A Lambert surface scatters with the *same* intensity into every outgoing direction, so the flux leaving it is

$$F_{\text{out}} = \int_0^{2\pi} \int_0^{\pi/2} I_{S,\text{scat}} \cos \theta \sin \theta \, d\theta \, d\phi = \pi I_{S,\text{scat}} = \pi p \cos \Theta I_{\text{inc}}(\nu).$$

3. **Energy conservation.** A *perfect* Lambert surface absorbs nothing, so $F_{\text{out}} = F_{\text{in}}$:

$$\pi p \cos \Theta I_{\text{inc}} = \pi \cos \Theta I_{\text{inc}} \implies \boxed{p = 1}$$

- The two factors of π — the hemispheric integral of $\cos \theta$ and the definition $I_{\text{inc}} \equiv F_{\text{inc}}/\pi$ — cancel, which is why p is simply the reflected fraction: a surface that reflects the fraction A has $p = A$.
- This $p = 1$ is what gave $A_g = 2/3$ for the Lambert sphere, and $A_S = A_g q = 1$: nothing absorbed.

The geometric albedo: general expression

Dividing numerator by denominator,

$$A_g(\nu) = \frac{\int_{-\pi/2}^{\pi/2} \int_{-\pi/2}^{\pi/2} I_{S,\text{scat}}(\theta, \phi, \nu) \cos \phi \cos^2 \theta \, d\theta \, d\phi}{\pi I_{\text{inc}}(\nu)}$$

- Something interesting: the denominator $\pi I_{\text{inc}}(\nu) = F_{\text{inc}}(\nu)$ is the incident stellar flux at the *substellar point*. So A_g may also be described as the ratio of the planet's scattered flux to the incident stellar flux at the substellar point.
- This is the form to use to compute A_g from the output of a *model atmosphere code*, which delivers the scattered intensity at the planetary surface as a function of location, $I_{S,\text{scat}}(\theta, \phi, \nu)$.
- The subscript “scat” means that only *scattered* radiation is counted, not radiation that has been absorbed and thermally reradiated; at visible wavelengths, for planets cooler than ~ 1500 K, the distinction is clean.

The geometric albedo: making the physics explicit

For a more conceptual understanding, relate the scattered intensity to the incoming intensity with the dimensionless $p(\Theta)$: the fraction of incident intensity scattered out of the planet into the angle Θ from the incident direction. All the physics of directional and multiple scattering inside the atmosphere is buried in p (it is *not* the single-scattering phase function P of Chapter 5):

$$I_{\text{S,scat}}(\theta, \phi, \nu) = I_{\text{inc}}(\theta', \phi', \nu) p(\Theta),$$

primes for the direction of incidence, no primes after scattering. For **uniform** incident stellar intensity, diluted away from the substellar point by the projection (Fig. 2.3),

$$I_{\text{inc}}(\theta', \phi', \nu) = \hat{\mathbf{n}} \cdot \hat{\mathbf{k}} I_{\text{inc}}(\nu) = \cos \Theta I_{\text{inc}}(\nu) = \cos \theta' \cos \phi' I_{\text{inc}}(\nu),$$

so that $I_{\text{S,scat}} = \cos \theta \cos \phi I_{\text{inc}}(\nu) p(\theta, \phi, \theta', \phi')$. Then $I_{\text{inc}}(\nu)$ cancels between numerator and denominator:

$$A_g(\nu) = \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} \int_{-\pi/2}^{\pi/2} p(\theta, \phi, \theta', \phi') \cos^2 \phi' \cos^3 \theta' d\theta' d\phi'$$

One factor $\cos \phi' \cos \theta'$ comes from the oblique illumination, the other $\cos \phi' \cos^2 \theta'$ from the flux projection and the solid angle toward the observer.

A_g of a Lambert sphere, and why A_g can exceed 1

For a **Lambert sphere** the intensity is scattered equally in all directions, $p(\Theta) = 1$, and

$$A_g^{\text{Lambert}} = \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} \int_{-\pi/2}^{\pi/2} \cos^2 \phi \cos^3 \theta \, d\theta \, d\phi = \frac{1}{\pi} \cdot \frac{\pi}{2} \cdot \frac{4}{3} = \boxed{\frac{2}{3}}$$

- ▶ $A_g = 2/3 < 1$ even though a Lambert sphere scatters *all* incident radiation and absorbs none. One third of the light is scattered out of the line of sight: the limb regions are illuminated obliquely *and* seen obliquely, so they contribute less than the corresponding parts of a flat disk.
- ▶ By the same definition, A_g can be greater than 1: it compares the planet not to a **a diffusing sphere** but to a **perfectly diffusing disk**. **Enceladus** has $A_g = 1.38$ at visible wavelengths (Verbiscer et al. 2007): fresh ice backscatters strongly, so at $\alpha = 0$ it returns more light toward the observer than a Lambert disk would. Rayleigh-scattering atmospheres also backscatter more than Lambert ($A_g = 3/4$ for a semi-infinite conservative Rayleigh atmosphere).
- ▶ Measurements of A_g have been elusive for exoplanets (upper limits from MOST, CoRoT, Kepler; a few detections such as Kepler-7b with $A_g \simeq 0.3$), but A_g remains the standard way to classify theoretical models.

The fraction scattered toward $\alpha = 0$ is A_g/π .

A_g compares the planet with a Lambert disk; it is *not* the fraction of the intercepted energy sent back toward the star. That fraction per steradian (exercise 3.5) follows in three steps.

1. **Power intercepted.** The planet presents its cross section to the star: $P_{\text{inc}} = \pi R_p^2 F_{\text{inc}}$.
2. **Power sent toward the observer.** By definition of A_g , with $\pi I_{\text{inc}} = F_{\text{inc}}$, the flux received at distance $D_{\oplus}$ at $\alpha = 0$ is $F_{\oplus} = A_g (R_p/D_{\oplus})^2 F_{\text{inc}}$. Multiplying by $D_{\oplus}^2$ turns this into power per unit solid angle leaving the planet in that direction: $(dP/d\Omega)_{\alpha=0} = D_{\oplus}^2 F_{\oplus} = A_g R_p^2 F_{\text{inc}}$.
3. **The ratio.** Dividing by the intercepted power,

$$\frac{1}{P_{\text{inc}}} \left(\frac{dP}{d\Omega} \right)_{\alpha=0} = \frac{A_g R_p^2 F_{\text{inc}}}{\pi R_p^2 F_{\text{inc}}} = \frac{A_g}{\pi} \text{ sr}^{-1}.$$

- It is a fraction *per steradian*. Energy conservation limits only its integral over all directions ($\leq A_B \leq 1$), which is why a strong backscatterer can have $A_g > 1$.
- Checks: a Lambert disk facing the star ($A_g = 1$) sends $1/\pi$ per sr along its normal, as the cosine law requires; a planet scattering everything isotropically sends $1/(4\pi)$ per sr, i.e. $A_g = 1/4$; a Lambert sphere ($A_g = 2/3$) sends $2/(3\pi)$, which is $8/3$ times the isotropic value — the same $8/3$ as the dayside factor f , from the same double-cos ϑ integral. Enceladus: $1.38/\pi \simeq 0.44$ per sr.

The Bond albedo A_B and the spherical albedo $A_S(\nu)$

- ▶ The **bond albedo** A_B is the fraction of incident stellar *energy* scattered back into space, including radiation into *all* directions and at *all* frequencies. By definition $A_B \leq 1$.
- ▶ The **spherical albedo** $A_S(\nu)$ is the same quantity at a specific frequency. The Bond albedo is the spherical albedo averaged over frequency *weighted by the incident stellar spectrum*:

$$A_B = \frac{\int_0^\infty A_S(\nu) F_{S,*}(\nu) d\nu}{\int_0^\infty F_{S,*}(\nu) d\nu}$$

(Seager writes $A_B = \int A_S d\nu$ as shorthand and states that the weighting by the incident radiation is implied; Heng's Eq. 2.10 gives the weighted form explicitly.)

- ▶ Consequently A_B depends on the *host star's spectrum*: the same planet has a different Bond albedo around a G star and around an M star, because an M star emits most of its light in the near-IR where the atmosphere may be dark.

To derive $A_S(\nu)$ we must account for the light scattered at *every* phase angle, not just at $\alpha = 0$. At each α only a portion of the hemisphere facing the observer is illuminated (next slide).

Scattered flux at phase angle α

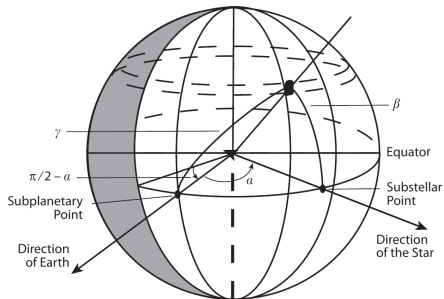


Figure 9: Coordinates for the phase-dependent scattered flux. θ is latitude from the intensity equator, ϕ longitude from the sub-Earth point, α the phase angle. After Sobolev (1975). Reproduced from Seager (2010), Fig. 3.6.

Measure longitude ϕ from the sub-Earth (subplanetary) point. The substellar point is then at longitude α , and a surface element at (θ, ϕ) is illuminated with the projection $\cos(\alpha - \phi) \cos \theta$ and seen with the projection $\cos \phi \cos \theta$. Only the longitudes $\alpha - \pi/2 < \phi < \pi/2$ are both lit and visible. The scattered flux toward the observer is

$$F_{S,\alpha}(\nu) = \int_{\alpha-\pi/2}^{\pi/2} \int_{-\pi/2}^{\pi/2} I_{\text{inc}}(\theta', \phi') p(\theta, \phi, \theta', \phi') \times \cos(\alpha - \phi') \cos \phi' \cos^3 \theta' d\theta' d\phi',$$

and for uniform incident intensity we write

$$F_{S,\alpha}(\nu) = I_{\text{inc}}(\nu) \Psi_{\alpha}(\nu),$$

where Ψ_{α} is the fraction of scattered radiation in the direction of the observer as a function of α . At $\alpha = 0$ it reduces to the geometric-albedo integral: $\Psi_{\alpha=0} = \pi A_g$.

The spherical albedo: integrating over all directions

To count the energy scattered into all directions, surround the planet with a sphere and use α as the polar angle and β as the azimuth (azimuthal symmetry assumed). Both are measured at the planet's center: α is the angle between the direction to the star and the direction to a point on the sphere — the phase angle of an observer there — and β the azimuth about the star direction. Each surface element reflects only into its own outward hemisphere, but the curved lit face as a whole sends light toward every α from 0 to π (with $\Psi_\alpha \rightarrow 0$ as $\alpha \rightarrow \pi$). The total scattered energy per unit time per frequency is

$$E_{\text{scat}}(\nu) = I_{\text{inc}}(\nu) \int_0^{2\pi} \int_0^\pi \Psi_\alpha(\nu) R_p^2 \sin \alpha \, d\alpha \, d\beta = 2\pi R_p^2 I_{\text{inc}}(\nu) \int_0^\pi \Psi_\alpha(\nu) \sin \alpha \, d\alpha.$$

The incident energy (Chapter 2) is $E_{\text{inc}}(\nu) = \pi R_p^2 \cdot \pi I_{\text{inc}}(\nu)$. Their ratio is the spherical albedo:

$$A_S(\nu) = \frac{E_{\text{scat}}(\nu)}{E_{\text{inc}}(\nu)} = \frac{2}{\pi} \int_0^\pi \Psi_\alpha(\nu) \sin \alpha \, d\alpha$$

- ▶ A_S needs Ψ_α at *all* phase angles. From Earth we cannot see all phases of most planets, so A_S (and A_B) are not directly measurable: we need a way to estimate them from the one thing we can hope to measure near full phase, A_g .
- ▶ Note that, in the factor $2/\pi$, the factor of 2 comes from the sphere-to-disk conversion, while the factor of π arises from the flux.

Relating A_S to A_g : the phase integral q

Using $\Psi_{\alpha=0}(v) = \pi A_g(v)$ (next slide), rewrite the spherical albedo as

$$A_S(v) = \frac{\pi A_g(v)}{\Psi_{\alpha=0}(v)} \cdot \frac{2}{\pi} \int_0^\pi \Psi_\alpha(v) \sin \alpha \, d\alpha \quad \Longrightarrow \quad \boxed{A_S(v) = A_g(v) q(v)}$$

with the **phase integral**

$$q(v) \equiv \frac{A_S(v)}{A_g(v)} = 2 \int_0^\pi \Phi_\alpha(v) \sin \alpha \, d\alpha, \quad \Phi_\alpha(v) \equiv \frac{\Psi_\alpha(v)}{\Psi_{\alpha=0}(v)}$$

where Φ_α is the **phase function**: the scattered flux versus phase angle normalized to full phase (this is the phase curve of the next lecture).

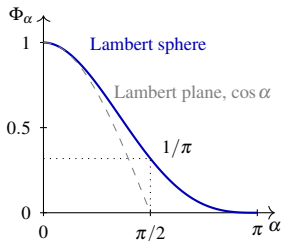
- ▶ A_S can be *estimated* from a measured A_g and a theoretical q . Heng writes the same thing as $A_S = A_g P_g$.
- ▶ **Lambert sphere:** $\Phi_\alpha = \frac{1}{\pi} [\sin \alpha + (\pi - \alpha) \cos \alpha]$ (derived on the next slide), so $q = 3/2$ and

$$A_g = \frac{2}{3} A_S, \quad A_S = A_B = 1, \quad A_g = \frac{2}{3}.$$

All light is scattered back to space, so the spherical albedo is 1; the geometric albedo counts only the light scattered toward the observer at $\alpha = 0$ and is less.

- ▶ **Rayleigh scattering:** $q = 4/3$; a Lambert *plane* (exercise 3.6): $\Phi_\alpha = \cos \alpha$. Real planets: $q \simeq 1.2$ – 1.7 .

The Lambert sphere: deriving Φ_α and q



Phase function of a Lambert sphere. At quadrature it is only $1/\pi$ as bright as full phase.

$$\Phi_0 = 1, \Phi_{\pi/2} = 1/\pi, \Phi_\pi = 0. \text{ With } \int_0^\pi \sin^2 \alpha \, d\alpha = \pi/2 \text{ and } \int_0^\pi (\pi - \alpha) \sin \alpha \cos \alpha \, d\alpha = \pi/4:$$

$$q = 2 \int_0^\pi \Phi_\alpha \sin \alpha \, d\alpha = \frac{2}{\pi} \left(\frac{\pi}{2} + \frac{\pi}{4} \right) = \frac{3}{2},$$

$$A_S = A_g q = \frac{2}{3} \cdot \frac{3}{2} = 1.$$

In general $\Psi_{\alpha=0} = \pi A_g$. Seager's Eq. (3.30) prints A_g/π , which would make her Eq. (3.31) inconsistent with Eq. (3.29).

For a perfect Lambert surface $p = 1$, and the integral for Ψ_α separates into a latitude and a longitude part:

$$\Psi_\alpha = \int_{-\pi/2}^{\pi/2} \cos^3 \theta \, d\theta \int_{\alpha-\pi/2}^{\pi/2} \cos(\alpha - \phi) \cos \phi \, d\phi.$$

- **Latitude:** $\int_{-\pi/2}^{\pi/2} \cos^3 \theta \, d\theta = \frac{4}{3}.$
- **Longitude:** with $\cos(\alpha - \phi) \cos \phi = \frac{1}{2} [\cos \alpha + \cos(\alpha - 2\phi)]$ over the lit and visible range of width $\pi - \alpha$,

$$\int_{\alpha-\pi/2}^{\pi/2} \cos(\alpha - \phi) \cos \phi \, d\phi = \frac{1}{2} \left[(\pi - \alpha) \cos \alpha + \left(-\frac{1}{2} \sin(\alpha - 2\phi) \right) \Big|_{\alpha-\pi/2}^{\pi/2} \right]$$

$$= \frac{1}{2} [\sin \alpha + (\pi - \alpha) \cos \alpha].$$

Hence

$$\Psi_\alpha = \frac{2}{3} [\sin \alpha + (\pi - \alpha) \cos \alpha], \quad \Psi_{\alpha=0} = \frac{2\pi}{3} = \pi A_g \Rightarrow A_g = \frac{2}{3},$$

$$\Phi_\alpha = \frac{1}{\pi} [\sin \alpha + (\pi - \alpha) \cos \alpha]$$

Bond versus geometric albedo for the giant planets

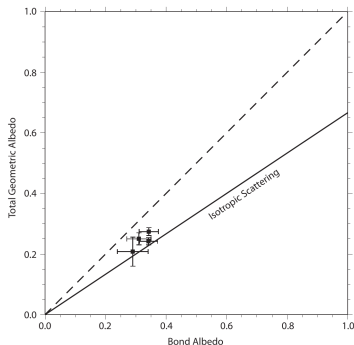


Figure 10: Total (wavelength-integrated) geometric albedo versus Bond albedo for a Lambert sphere (solid, $A_g/A_B = 2/3$) and for Uranus, Neptune, Saturn, Jupiter (points, in order of increasing A_B). The dashed line is $A_g = A_B$. Adapted from Rowe et al. (2006); reproduced from Seager (2010), Fig. 3.7.

- ▶ All four giants lie *above* the Lambert (isotropic scattering) line: their atmospheres backscatter more than a Lambert sphere, so $A_g/A_B > 2/3$ ($q < 3/2$).
- ▶ The wedge between the Lambert line and the line $A_g = A_B$ bounds the photometric properties of most spherical bodies with deep atmospheres (Rayleigh scattering, clouds, dust). **A planet with deep atmosphere must lie to the right of the dashed line.**
- ▶ Practical use: a measured (or upper-limit) A_g of an exoplanet, e.g. the MOST limit $A_g < 0.25$ for HD 209458b, can be converted to a range of A_B , hence to a range of T_{eq} .

The apparent albedo: the one an exoplanet can give us

The geometric, spherical, and Bond albedos are unsatisfactory for exoplanets because *none can be measured*:

- ▶ A_g is defined at full phase ($\alpha = 0$), when the planet is behind the star and unobservable (for transiting planets the eclipse depth measured just before/after eclipse is the practical substitute, at a small α).
- ▶ A_S and A_B require all phase angles. With direct imaging, phases where the planet projects too close to the star are inaccessible, and the planet does not go through all phases at all unless the orbit is edge-on.

Apparent albedo. The ratio of the scattered flux emerging from the planet toward the observer to the scattered flux from a *perfectly reflecting Lambert sphere* of the same size, at the same phase angle, at the same distance:

$$A_{\text{app}}(\alpha, \nu) = \frac{F_{S,\alpha}(\nu)}{F_{S,\alpha}^{\text{Lambert}}(\nu)} = \frac{A_g(\nu) \Phi_\alpha(\nu)}{\frac{2}{3} \Phi_\alpha^{\text{Lambert}}} \quad (\text{exercise 3.9}).$$

- ▶ It is the planet's reflectance in the direction of the observer at a given time, a function of phase angle and frequency.
- ▶ It is the only albedo that can be derived from a *single* observation of a planet of known size — a directly imaged planet at known separation and phase.

Take-home messages

Temperatures. Energy balance gives $T_{\text{eq}} = T_{\text{eff},*}(R_*/a)^{1/2}[f'(1 - A_B)]^{1/4}$, with $f' = 1/4$ for uniform redistribution and $2/3$ for an instantaneously reradiating dayside. T_{eff} matches the total emitted flux and exceeds T_{eq} when the interior supplies heat; $T_b(\nu)$ matches the flux at one frequency and is the only temperature measurable for an exoplanet.

Albedos. A_g compares the full-phase flux with a Lambert disk (Lambert sphere: $2/3$; A_g can exceed 1); $A_S = A_g q$ with the phase integral q ($3/2$ for Lambert); A_B is A_S averaged over the stellar spectrum and is the one that enters the energy balance.

Flux ratios. $F_{\oplus,p}/F_{\oplus,*} = F_{S,p}R_p^2/F_{S,*}R_*^2$. Thermal: $(R_p/a)^2(f/4)(1 - A_B)$ bolometrically, $(T_{b,p}/T_{b,*})(R_p/R_*)^2$ in the Rayleigh–Jeans regime. Reflected at full phase: $A_g R_p^2/a^2$ (geometric, not Bond, albedo; $g = 4A_g/A_S$). Transmission: $2R_p A_H/R_*^2 \sim 10^{-3}$ for hot Jupiters.

Phase curves. $A_g(R_p/a)^2\Phi_\alpha$ with the Lambert phase function $\Phi_\alpha = [\sin \alpha + (\pi - \alpha) \cos \alpha]/\pi$; thermal phase curves follow the same law when the absorbed energy is reradiated instantaneously, and depart from it when winds redistribute heat.

- ▶ Seager, S. 2010, *Exoplanet Atmospheres: Physical Processes* (Princeton University Press), Chapter 3.
- ▶ Heng, K. 2017, *Exoplanetary Atmospheres: Theoretical Concepts and Foundations* (Princeton University Press), Section 2.2.3 (albedos).
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- ▶ Cox, A. N. 2000, *Allen's Astrophysical Quantities*, 4th ed. (Springer) — Solar System temperatures, albedos, and phase-curve formulae.
- ▶ Verbiscer, A., et al. 2007, *Science* **315**, 815 (Enceladus, $A_g = 1.38$); Rowe, J., et al. 2006, *ApJ* **646**, 1241 (MOST albedo limit for HD 209458b); Conrath, B. J., Hanel, R. A., and Samuelson, R. E. 1989, in *Origin and Evolution of Planetary and Satellite Atmospheres*, 513 (giant-planet albedos).
- ▶ Pearl, J. C. and Christensen, P. R. 1997, *JGR* **102**, 10875 (Earth's mid-IR spectrum).
- ▶ Knutson, H. A., et al. 2007, *Nature* **447**, 183; Zellem, R. T., et al. 2014, *ApJ* **790**, 53 (infrared phase curves).